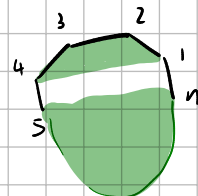
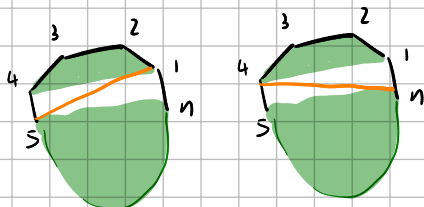


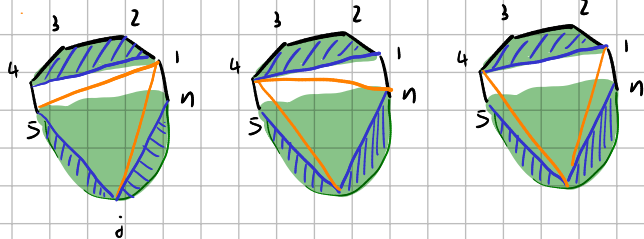
3-zeros



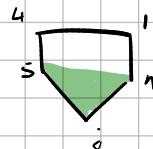
1 BP



2 BP



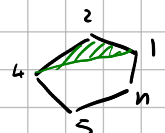
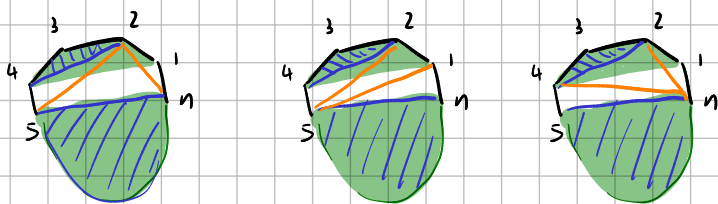
same as 2 BP in 5-gon for 1-zero bound at 1



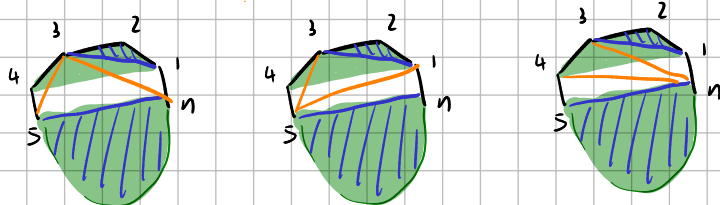
$$\Rightarrow \frac{1}{x_{15} x_{45}} + \frac{1}{x_{4n} x_{4i}} + \frac{1}{x_{1i} x_{4i}} = \frac{x_{4n} x_{4i} + x_{15} x_{4i} + x_{15} x_{4n}}{x_{15} x_{45} x_{4n} x_{4i} x_{1i} x_{4i}}$$

impose constraint!

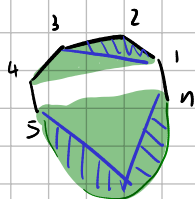
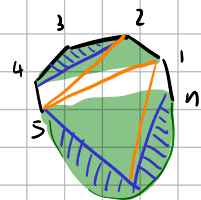
$$-x_{15} (x_{1i} - x_{45}) + x_{15} x_{4i} - x_{1i}^2 = 0$$



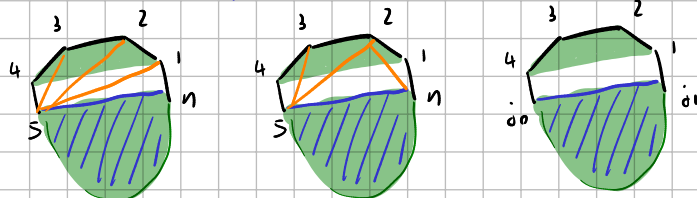
5-gon for 2-zero



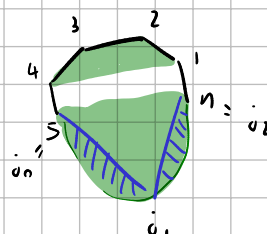
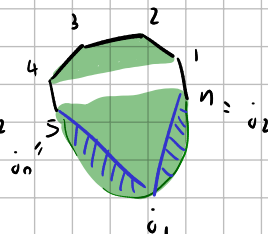
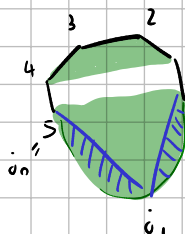
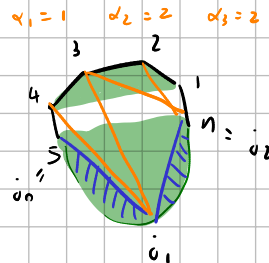
3 BP



new



4 BP



$$\sum_{\alpha_1=0}^{s+1} \sum_{\alpha_2=\alpha_1}^{s+1} \sum_{\alpha_3=\alpha_2}^{s+1} \frac{1}{\prod_{a=1}^{\alpha_1} X_{4ja}} \frac{1}{\prod_{a=\alpha_1}^{\alpha_2} X_{3ja}} \frac{1}{\prod_{a=\alpha_2}^{\alpha_3} X_{2ja}} \frac{1}{\prod_{a=\alpha_3}^s X_{1ja}}$$

(siehe as 2-zero pattern but with an more sum)

$s = \# \text{ BP} - 3$

pattern

count # of projections starting at 1, 2, 3, 4

Gamma denominator:

$$\sum_{\alpha_1=0}^{s+1} \sum_{\alpha_2=\alpha_1}^{s+1} \sum_{\alpha_3=\alpha_2}^{s+1} \prod_{a=0}^{\alpha_3-1} X_{1ja} \prod_{a=0}^{\alpha_2-1} X_{2ja} \prod_{a=\alpha_{s+1}}^{s+1} X_{2ja} \prod_{a=0}^{\alpha_1-1} X_{3ja} \prod_{a=\alpha_{s+1}}^{s+1} X_{3ja} \prod_{a=\alpha_{s+1}}^{s+1} X_{4ja}$$

trick: insert the constraint but add extra variable

$X_{1n} = X_{1, s+1}$ (with $j, s+1 = n$)

constraint bar line $X_{4n} = X_{1n} - X_{1s}$

$$\rightarrow \begin{cases} X_{4ja} = X_{1ja} - X_{1jo} \\ X_{2ja} = X_{1ja} - X_{1jo} + X_{2jo} \\ X_{3ja} = X_{1ja} - X_{1jo} + X_{3jo} \end{cases} \quad a = 0, \dots, s+1$$

note: actual real constraint also not have $X_{1, s+1} = 0$, which we insert later.

$$\rightarrow \sum_{\alpha_1=0}^{s+1} \sum_{\alpha_2=\alpha_1}^{s+1} \sum_{\alpha_3=\alpha_2}^{s+1} \prod_{a=0}^{\alpha_3-1} X_{1ja} \prod_{a=0}^{\alpha_2-1} (X_{1ja} - X_{1jo} + X_{2jo}) \prod_{a=\alpha_{s+1}}^{s+1} (X_{1ja} - X_{1jo} + X_{2jo})$$

$$\prod_{a=0}^{\alpha_1-1} (X_{1ja} - X_{1jo} + X_{3jo}) \prod_{a=\alpha_{s+1}}^{s+1} (X_{1ja} - X_{1jo} + X_{3jo}) \prod_{a=\alpha_{s+1}}^{s+1} (X_{1ja} - X_{1jo})$$

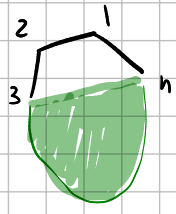
proof by induction, and it extends to any k !

$$= \prod_{a=1}^{s+1} X_{1ja} (X_{1ja} - X_{1jo} + X_{2jo}) (X_{1ja} - X_{1jo} + X_{3jo}) \propto X_{1, s+1}$$

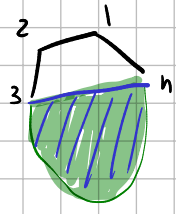
when we set $X_{1, s+1} = 0$ the whole thing vanishes

idea: focus on $(n-3)$ -zeros for $n-3$ boundary propagators and see what happens
and possibly use induction

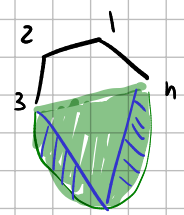
ex: 1-zero



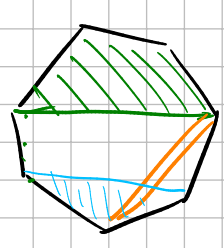
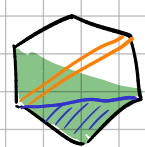
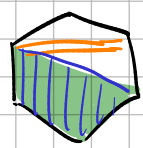
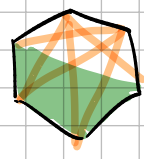
1BP



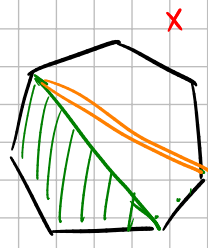
2BP



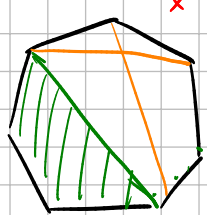
conjecture: if we fit 2 boundary propagators in an n -gon, they only span a S -gon if they don't cross
FALSE



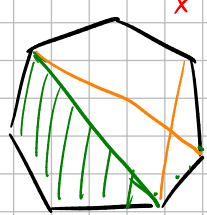
+



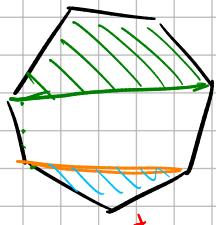
x



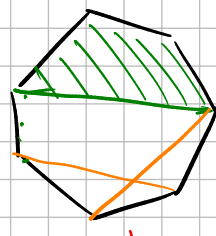
x



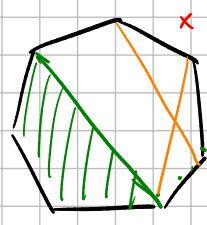
x



+



+



x

1BP 1-zero

$$\sum_{i=3}^{n-1} \frac{a_{1i}}{X_{1i}} + \sum_{i=4}^n \frac{a_{2i}}{X_{2i}} \propto \sum_{i=3}^{n-1} a_{1i} \prod_{\substack{l=3 \\ l \neq i}}^{n-1} X_{1l} \prod_{s=4}^n X_{2s} + \sum_{i=4}^n a_{2i} \prod_{\substack{s=4 \\ s \neq i}}^n X_{2s} \prod_{l=3}^{n-1} X_{1l}$$

impose constraint

$$\sum_{i=3}^{n-1} a_{1i} \prod_{\substack{l=3 \\ l \neq i}}^{n-1} X_{1l} \prod_{s=4}^n (X_{1s} - X_{13}) + \sum_{i=4}^n a_{2i} \prod_{\substack{s=4 \\ s \neq i}}^n (X_{1s} - X_{13}) \prod_{l=3}^{n-1} X_{1l}$$

set X_{1i} to zero:

$$a_{1i} = 0$$